PRACTICAL NO. 4: MULTIPLE LINEAR REGRESSION

a. Establish the linear relationship of value of house on salary of employee and number of bed room in the house.

Descriptive Statistics

	Mean	Std. Deviation	N
Value of the house in thousand of dollar	256.69	61.838	150
Salary in thousand of dollar	72.15	21.052	150
Number of bedrooms in the house	3.17	.749	150

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	289777.459	2	144888.730	76.070	<.001[b]
	Residual	279986.434	147	1904.670		
	Total	569763.893	149			

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.713[a]	.509	.502	43.643

Correlations

		Value of the house in thousand of dollar	Salary in thousand of dollar	Number of bedrooms in the house
Pearson Correlation	Value of the house in thousand of dollar	1.000	.557	.685
	Salary in thousand of dollar	.557	1.000	.575
	Number of bedrooms in the house	.685	.575	1.000
Sig. (1-tailed)	Value of the house in thousand of dollar	.	<.001	<.001
	Salary in thousand of dollar	.000	.	.000
	Number of bedrooms in the house	.000	.000	.
N	Value of the house in thousand of dollar	150	150	150
	Salary in thousand of dollar	150	150	150
	Number of bedrooms in the house	150	150	150

Interpretation: The regression equation is $Y = 62.267 + 0.717X_1 + 44.972X_2$

Where Y = value of house, X_1 = salary of the employee and X_2 = No of bedrooms.

i. What is the value of the house can we expect with the salary of the employee is 90 thousand dollar and having 5 bed rooms.

Solution: Here, $X_1 = 90$ thousand and $X_2 = 5$ bedrooms.

Now $Y = 62.267 + 0.717(90) + 44.972(5) = 351.66$

Hence, the expected house value is approximately 351.66 thousand dollars.

ii. Test the significance of regression coefficients and overall fit of the model.

Solution:

Coefficients[a]

Model		Unstandardized Coefficients B	Unstandardized Coefficients Std. Error	Standardized Coefficients Beta	t	Sig.	Correlations Zero-order	Correlations Partial	Correlations Part
1	(Constant)	62.267	16.161		3.853	<.001			
	Salary in thousand of dollar	.717	.208	.244	3.454	<.001	.557	.274	.200
	Number of bedrooms in the house	44.972	5.836	.544	7.706	<.001	.685	.536	.446

a. Dependent Variable: Value of the house in thousand of dollar

Interpretation: Since p-value $< 0.001 < 0.05$. So, we reject H_0 and the model is significant.
Hence, there is significant evidence that salary and number of bedrooms influence the value of the house.

iii. Determine simple, partial correlation coefficients and coefficient of determination and interpret it.

Solution:

Coefficients[a]

Model		Unstandardized Coefficients B	Unstandardized Coefficients Std. Error	Standardized Coefficients Beta	t	Sig.	Correlations Zero-order	Correlations Partial	Correlations Part
1	(Constant)	62.267	16.161		3.853	<.001			
	Salary in thousand of dollar	.717	.208	.244	3.454	<.001	.557	.274	.200
	Number of bedrooms in the house	44.972	5.836	.544	7.706	<.001	.685	.536	.446

a. Dependent Variable: Value of the house in thousand of dollar

Interpretation: Simple correlation shows a moderate positive relationship between salary and house value ($r = 0.557$) and a strong positive relationship between bedrooms and house value ($r = 0.685$).

After controlling for the other variable, partial correlation reduces to weak for salary ($r = 0.274$) and moderate for bedrooms ($r = 0.536$).

Part correlation indicates that salary uniquely explains 4.0% of the variance in house value, while bedrooms uniquely explains 19.9%, confirming that number of bedrooms is a stronger unique predictor.

Thus, salary and number of bedrooms are significant factors influencing house value, with number of bedrooms showing the strongest relationship.

b. Determine the linear relationship of salary of employee and number of years in the office, education level, Sick days taken during the previous year.

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	Change Statistics				
					R Square Change	F Change	df1	df2	Sig. F Change
1	.675[a]	.456	.444	15.693	.456	40.717	3	146	<.001

Descriptive Statistics

	Mean	Std. Deviation	N
Salary in thousand of dollar	72.15	21.052	150
Number of year with the company	3.02	1.190	150
Education of the employee	3.55	1.020	150
Sick days taken during the previous year	8.27	4.557	150

Correlations

		Salary in thousand of dollar	Number of year with the company	Education of the employee	Sick days taken during the previous year
Pearson Correlation	Salary in thousand of dollar	1.000	.626	.390	.026
	Number of year with the company	.626	1.000	.234	.135
	Education of the employee	.390	.234	1.000	-.089
	Sick days taken during the previous year	.026	.135	-.089	1.000
Sig. (1-tailed)	Salary in thousand of dollar	.	<.001	<.001	.375
	Number of year with the company	.000	.	.002	.050
	Education of the employee	.000	.002	.	.139
	Sick days taken during the previous year	.375	.050	.139	.
N	Salary in thousand of dollar	150	150	150	150
	Number of year with the company	150	150	150	150
	Education of the employee	150	150	150	150
	Sick days taken during the previous year	150	150	150	150

Interpretation: The regression equation is $Y = 24.143+10.099(X_1)+5.231(X_2)-0.131(X_3)$

Where Y = Salary, X_1 = No of Years with the company, X_2 = Education of employee and X_3 = Sick days taken during the previous year

i. Test the significance of regression coefficients and overall fit of the model.

Solution:

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.	95.0% Confidence Interval for B		Correlations		
		B	Std. Error	Beta			Lower Bound	Upper Bound	Zero-order	Partial	Part
1	(Constant)	24.143	5.717		4.223	<.001	12.845	35.441			
	Number of year with the company	10.099	1.126	.571	8.969	<.001	7.874	12.325	.626	.596	.548
	Education of the employee	5.231	1.306	.253	4.004	<.001	2.649	7.813	.390	.315	.244
	Sick days taken during the previous year	-.131	.287	-.028	-.456	.649	-.698	.436	.026	-.038	-.028

a. Dependent Variable: Salary in thousand of dollar

Interpretation: Since p-value < 0.001 < 0.05. Hence, we reject H_0. So, the model is Significant.

There is significant evidence that years in the company, education level, and sick days collectively influence the salary of employees.

ii. Test the partial correlation coefficient and zero order correlation coeff. and interpret it.

Solution:

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.	95.0% Confidence Interval for B		Correlations		
		B	Std. Error	Beta			Lower Bound	Upper Bound	Zero-order	Partial	Part
1	(Constant)	24.143	5.717		4.223	<.001	12.845	35.441			
	Number of year with the company	10.099	1.126	.571	8.969	<.001	7.874	12.325	.626	.596	.548
	Education of the employee	5.231	1.306	.253	4.004	<.001	2.649	7.813	.390	.315	.244
	Sick days taken during the previous year	-.131	.287	-.028	-.456	.649	-.698	.436	.026	-.038	-.028

a. Dependent Variable: Salary in thousand of dollar

Interpretation: Zero-order correlation: Years with company ($r = 0.626$) and education level ($r = 0.390$) show moderate to strong positive correlation with salary. Sick days ($r = 0.026$) show no correlation with salary.

Partial correlation (after controlling for other predictors): Years ($r = 0.596$) and education ($r = 0.315$) still show positive correlation with salary. Sick days ($r = -0.038$) show negligible correlation.

Part correlation (unique contribution): Years uniquely explains 30.0% of variance in salary. Education uniquely explains 5.95% of variance. Sick days explains only 0.08%, which is negligible.

Thus, years in the company and education level are significantly related to salary, whereas sick days taken during the previous year are not significantly related to salary.

PRACTICAL NO. 5: NON-PARAMETRIC TEST

a. Run test: The sequence of 150 employees (male and female) is given in the data set. Test at 5% level of significance that whether the sequence of male and female is in random order?

Hypothesis:

H_0: Sequence of male and female employees is random
v/s H_1: Sequence is not random

Descriptive Statistics

	N	Mean	Std. Deviation	Minimum	Maximum
Sex of employee	150	.60	.492	0	1

Runs Test

	Sex of employee
Test Value[a]	1
Cases < Test Value	60
Cases >= Test Value	90
Total Cases	150
Number of Runs	73
Z	.000
Asymp. Sig. (2-tailed)	1.000

Interpretation: Here, no of runs = 73. Since p-value = 1.000 > 0.05. Hence, we do not reject H_0. There is strong evidence to say that the sample is in random order.

b. Binomial test: Out of 150 employees, use binomial test to test the hypothesis that male and female are equally employed.

Hypothesis:

H_0: P = 0.5

v/s H_1: P ≠ 0.5

Descriptive Statistics

	N	Mean	Std. Deviation	Minimum	Maximum
Sex of employee	150	.60	.492	0	1

Binomial Test

		Category	N	Observed Prop.	Test Prop.	Exact Sig. (2-tailed)
Sex of employee	Group 1	<= .5	60	.40	.50	.018
	Group 2	> .5	90	.60		
	Total		150	1.00		

Interpretation: Since p-value = 0.018 < 0.05, we reject H_0.

There is significant evidence that males and females are not equally employed.

The proportion of male employees (60%) is significantly higher than female employees (40%).

c. **Kolmogorov – Smirnov test:** Given that random sample of 150 employee's value of houses (in thousands of rupees). Apply the Kolmogorov- Smirnov test for testing that value of the house is uniformly distributed.

Hypothesis:

H_0: Samples come from population with equal distribution

v/s H_1: Samples come from population not with equal distribution

One-Sample Kolmogorov-Smirnov Test

			Value of the house in thousand of dollar
N			150
Uniform Parameters[a,b]	Minimum		120
	Maximum		400
Most Extreme Differences	Absolute		.172
	Positive		.172
	Negative		-.114
Test Statistic			.172
Monte Carlo Sig. (2-tailed)[c]	Sig.		<.001
	99% Confidence Interval	Lower Bound	.000
		Upper Bound	.001

a. Test distribution is Uniform.

b. Calculated from data.

c. Lilliefors' method based on 10000 Monte Carlo samples with starting seed 2000000.

Interpretation: Since p-value < 0.05, we reject the H_0.

There is significant evidence that the value of houses does not follow a uniform distribution. The observed distribution differs significantly from a uniform distribution.

d. Mann Whitney U test: Test the hypothesis of no difference between the salary of the employee of swimming pool in house or not, using Mann Whitney U test for the sample data of 150 employees of the company.

Hypothesis:

H_0: $Md_1 = Md_2$

v/s H_1: $Md_1 \neq Md_2$

Mann-Whitney Test

Ranks

	Swimming pool in the house	N	Mean Rank	Sum of Ranks
Salary in thousand of dollar	No	68	50.82	3455.50
	Yes	82	95.97	7869.50
	Total	150		

Test Statistics[a]

	Salary in thousand of dollar
Mann-Whitney U	1109.500
Wilcoxon W	3455.500
Z	-6.342
Asymp. Sig. (2-tailed)	<.001

a. Grouping Variable: Swimming pool in the house

Interpretation: Since p-value < 0.05, we reject H_0.

There is significant evidence of a difference in salary between employees with and without a swimming pool. Employees with a swimming pool have significantly higher salary than those without.

e. Chi square test: use Chi square test of independence to determine an association between education and number of years in the company.

Hypothesis:

H_0: $O_i = E_i$

v/s H_1: $O_i \neq E_i$

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	39.153[a]	16	.001
Likelihood Ratio	33.705	16	.006
Linear-by-Linear Association	8.166	1	.004
N of Valid Cases	150		

a. 12 cells (48.0%) have expected count less than 5. The minimum expected count is .68.

Interpretation: Since p-value = 0.001 < 0.05, we reject H_0.

There is significant evidence of an association between education level and number of years in the company. The two variables are not independent; education level and tenure are related.

f. Median test: Use the median test to determine whether the distance (in miles) from house to work for male and female are same in the company.

Hypothesis:

H_0: Median distance to work for male and female employees is the same

H_1: Medians are different

Frequencies

		Sex of employee	
		Female	Male
Distance in miles from house to work	> Median	31	42
	<= Median	29	48

Test Statistics[a]

		Distance in miles from house to work
N		150
Median		15.00
Chi-Square		.360
df		1
Asymp. Sig.		.548
Yates' Continuity Correction	Chi-Square	.188
	df	1
	Asymp. Sig.	.665

a. Grouping Variable: Sex of employee

Interpretation: Since p-value = 0.548 > 0.05, we do not reject H_0.

p-value = 0.548 > 0.05 → Do not reject H0.
Median commuting distance is the same for male and female employees.